

Defects

In

Solids



96
C.L. → # what are the defects in solids? describe them.

The term defect denotes a state of the lattice in which the atomic arrangement has departed from regularity. There are different kinds of defects;

1. Point defects (i)
2. Line defects (ii)
3. plane defects (iii)
4. Electronic imperfections
5. Excitation states of crystals
6. Transient imperfections.

The description of these defects are as follows:-

1. Point defects:-

If the deviation from regularity is confined to very small region of only about a few lattice constants it is called a point defect. It is of four types:-

- i) Interstitial atom  (i)
- ii) Schottky defect or vacancy  (ii)
- iii) Frenkel defect  (iii)
- iv) substitutional impurity atom  (iv)

2. Line defects:-

The example of this defect is dislocation. It is a region of strain in the crystal in which, part of the lattice has undergone a shearing strain equal to one lattice vector called a Burgers vector. It is of two types,

- i) Edge dislocation
- ii) screw dislocation (iii)

3. Plane defects:-

These defects have an extension in area and are confined to a small region. There are three types of plane defects.

- i) Lineage boundary
- ii) Grain boundary
- iii) Stacking fault.

4. Electronic imperfections:-

Errors in charge distribution or energy in solids are called electronic defects. These defects are responsible for important electrical and magnetic properties.

5. Excitation states of crystals:-

These are imperfections as these cause deviations from perfect crystal symmetry. Some examples are

- i) Phonons and magnons.
- ii) conduction electrons and holes
- iii) Excitons
- iv) Quantised plasma waves.

6. Transient imperfections:-

These imperfections are introduced into the crystal from external sources, i.e.,

- i) photons
- ii) High energy charged particles like electrons, protons, mesons and ions.
- iii) High energy uncharged particles such as neutrons and neutral atoms.

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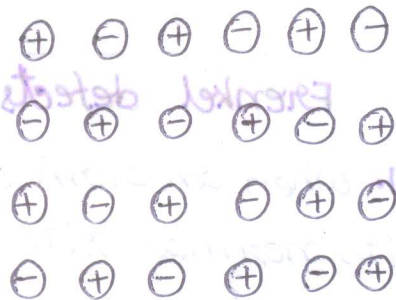
Find the difference between schottky and Frenkel defects.

⇒ The differences between schottky and Frenkel defects are given below:

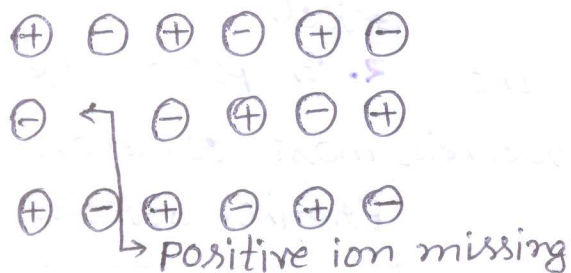
Schottky defects	Frenkel defects
1. This is a defect in which an atom or ion is found to be missing from its correct position.	1. When an atom or ion leaves its normal site and is found to occupy another position in the interstice, we get a Frenkel defect.
2. In pure alkali halides the most common lattice vacancies are schottky defects.	2. In pure silver halides the most common vacancies are Frenkel defects.
3. The production of schottky defects lowers the density of the crystal because the volume is increased with no increasing of mass.	3. The production of Frenkel defects does not change the volume of crystal, thus the density remains unchanged.
4. After various degrees of dissociation the two components of schottky defects are not independent of each other.	4. After various degrees of dissociation the two components of the Frenkel defects are independent of each other.
5. There are the individual positive and negative or both types of ion vacancies. Therefore, schottky defects only vacancy.	5. In this case two imperfection is created; an interstitial atom and a vacancy.

Schottky defects

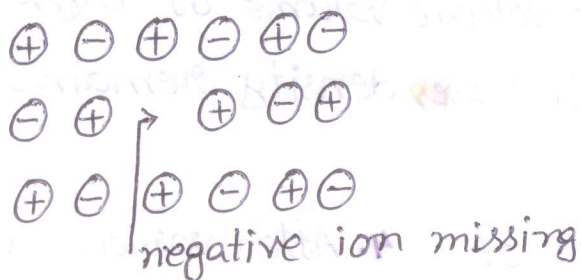
a) crystal structure



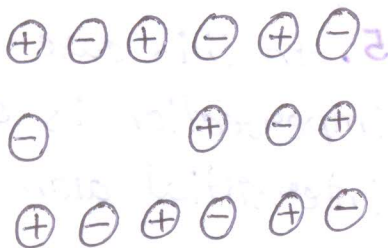
b) Schottky defect



c) Negative ion missing schottky defect



d) pair of ion missing schottky defect

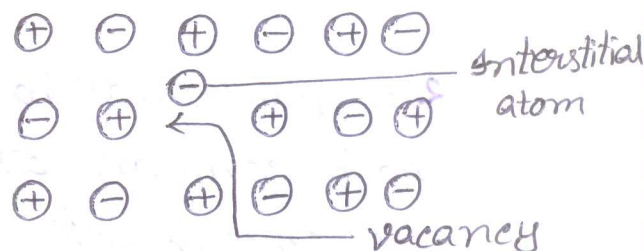


Frenkel defects

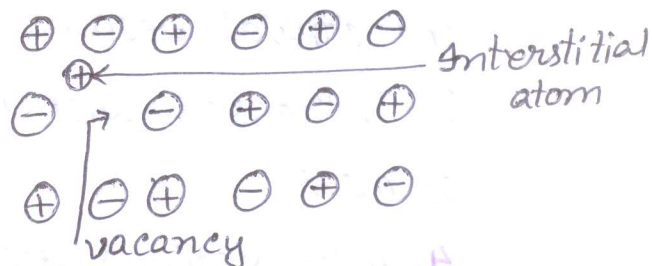
a) crystal structure



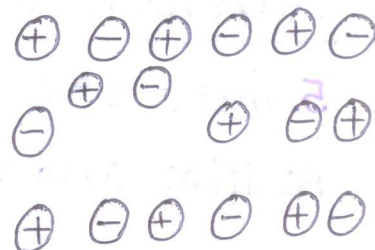
b) Frenkel defect



c) positive ion displacement and vacancy Frenkel defect.



d) pair of ion displacement and vacancy Frenkel defect



96.97.00 # Deduce the following expression for the equilibrium concentration of vacancies:-

$$n \cong N \exp \left(- \frac{E_v}{k_B T} \right)$$

where the terms carry usual meaning.

OR

calculate the number densities of the positive and negative defects in a system maintaining charge neutrality.

⇒ At a finite temperature T , the equilibrium condition of a crystal is the state of minimum free energy,

$$F = U - TS \quad \text{--- (1)}$$

Let E_v is the energy required to take an atom from a lattice site inside the crystal to a lattice site on the surface.

Then, $U = nE_v$ is the increase in energy associated with the production of n isolated vacant sites.

The total number of ways in which we can pick n atoms from a crystal containing N -atoms is,

$$W = \frac{N(N-1) \dots (N-n+1)}{n!} = \frac{N!}{(N-n)! n!} \quad \text{--- (2)}$$

The entropy is given by Boltzmann's statistical relation,

$$S = k_B \log W \quad \text{--- (3)}$$

From equations (2) and (3), we have,

$$S = k_B \log \frac{N!}{(N-n)! n!} \quad \text{--- (4)}$$

The configurational entropy due to production of n schottky defects produces a change in the free energy is,

$$F = U - TS$$

$$= nE_v - k_B T \log \frac{N!}{(N-n)!n!} \text{ --- (5)}$$

The logarithmic term is expanded by Stirling approximation $\log x! = x \log x - x$; for $x \gg 1$

Hence,

$$\begin{aligned} \log \left[\frac{N!}{(N-n)!n!} \right] &= \log N! - \log(N-n)! - \log n! \\ &= N \log N - N - (N-n) \log(N-n) + (N-n) - n \log n + n \\ &= N \log N - (N-n) \log(N-n) - n \log n \text{ --- (6)} \end{aligned}$$

Then, equⁿ (5) becomes,

$$F = nE_v - k_B T [N \log N - (N-n) \log(N-n) - n \log n] \text{ --- (7)}$$

Differentiating equⁿ (7) with respect to n ,

$$\begin{aligned} \frac{\partial F}{\partial n} &= E_v - k_B T \left[0 - \frac{(N-n)}{(N-n)} (-1) - (-1) \log(N-n) - 1 - \log n \right] \\ &= E_v - k_B T [1 + \log(N-n) - 1 - \log n] \\ &= E_v - k_B T \log \frac{N-n}{n} \text{ --- (8)} \end{aligned}$$

when equilibrium is attained at a given temperature T , the free energy is constant.

$$\text{i.e., } \frac{\partial F}{\partial n} = 0$$

$$\Rightarrow E_v - k_B T \log \left(\frac{N-n}{n} \right) = 0$$

$$\Rightarrow E_v = k_B T \log \left(\frac{N-n}{n} \right)$$

$$\Rightarrow \log \left(\frac{N-n}{n} \right) = \frac{E_v}{k_B T}$$

$$\Rightarrow \frac{N-n}{n} = \exp\left(\frac{E_v}{k_B T}\right) \text{ --- --- --- --- --- (9)}$$

The number of schottky defects in a crystal is very much smaller than the number of atoms and so far $N \gg n$.

we have $(N-n) \approx N$

Then, $\frac{N}{n} \approx \exp\left(\frac{E_v}{k_B T}\right)$

$$\therefore n \approx N \exp\left(-\frac{E_v}{k_B T}\right) \text{ --- --- --- --- --- (10)}$$

Equⁿ (10) is the required expression for the equilibrium and also the required number densities.

Q.2 # show that, the concentration of schottky defects in an ionic crystal is given by,

$$n \approx N \exp\left(-\frac{E_p}{2k_B T}\right)$$

where the terms carry usual meaning.

⇒ According to thermodynamics, the equilibrium concentration of a solid, under low external pressure at a temperature T is determined by the minimum value of the Helmholtz free energy,

$$F = U - TS \text{ --- --- --- --- --- (1)}$$

where, U - is the internal energy of the crystal and S is its entropy.

The energy E_p of a vacancy pair is the energy required to take a molecule (anion and cation) from the interior of the crystal to its surface.

If n be the number of vacancy pairs, the total increase in the energy due to the production of the vacancies is given by,

$$U = nE_p \quad \text{--- (2)}$$

The entropy is given by the Boltzmann's statistical relation,

$$S = k_B \log w \quad \text{--- (3)}$$

Let, the total number of atoms in the crystal be N . The different ways in which each kind of ion can be picked is thus given by,

$$W = \frac{N(N-1)(N-2) \dots (N-n+1)}{n!} = \frac{N!}{(N-n)! n!} \quad \text{--- (4)}$$

Taking into consideration the number of ways of removing the components of the pair which are equal in number each factor must be squared. Hence,

$$W = \left[\frac{N!}{(N-n)! n!} \right]^2$$

Now, from equⁿ (3),

$$S = k_B \log \left[\frac{N!}{(N-n)! n!} \right]^2 \quad \text{--- (5)}$$

The configurational entropy due to the production of n Schottky defects produces a change in the Helmholtz free energy of the vacancies which is given by substituting equⁿs (2) and (5) into equⁿ (1),

$$F = nE_p - k_B T \log \left[\frac{N!}{(N-n)! n!} \right]^2 \quad \text{--- (6)}$$

The logarithmic term is expanded by stirling's formula,

$$\log x! \approx x \log x - x$$

Hence,

$$\begin{aligned} \log \left[\frac{N!}{(N-n)! n!} \right]^2 &\approx 2 [\log N! - \log (N-n)! - \log n!] \\ &= 2 [N \log N - N - (N-n) \log (N-n) + (N-n) - n \log n + n] \\ &= 2 [N \log N - (N-n) \log (N-n) - n \log n] \quad \text{--- (7)} \end{aligned}$$

Putting this value in equⁿ (6), we have,

$$F = nE_p - 2k_B T [N \log N - (N-n) \log (N-n) - n \log n] \quad \text{--- (8)}$$

Differentiating equⁿ (8) with respect to n , we get

$$\begin{aligned} \frac{\partial F}{\partial n} &= E_p - 2k_B T [0 + 1 + \log (N-n) - 1 - \log n] \\ &= E_p - 2k_B T \log \frac{(N-n)}{n} \end{aligned}$$

when equilibrium is attained, the free energy is constant,

so,
$$\frac{\partial F}{\partial n} = 0 = E_p - 2k_B T \log \frac{N-n}{n}$$

$$\Rightarrow E_p = 2k_B T \log \frac{N-n}{n}$$

$$\Rightarrow \log \frac{N-n}{n} = \frac{E_p}{2k_B T}$$

$$\Rightarrow \frac{N-n}{n} = \exp \left(\frac{E_p}{2k_B T} \right)$$

The number of schottky defects in a crystal is very much smaller than the number of atoms and $n \ll N$, we have, $(N-n) \approx N$. Thus the relative density of vacancies is,

$$\frac{n}{N} = \exp \left(- \frac{E_p}{2k_B T} \right)$$

The number of schottky defects is given by,

$$n \approx N \exp \left(- \frac{E_p}{2k_B T} \right).$$

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Show that, the number of Frenkel defects per unit volume in a crystal is given by,

$$n \cong (NN')^{1/2} \exp\left(-\frac{E_i}{2k_B T}\right)$$

where the symbols have their usual meaning.

⇒ This is one type of lattice vacancy. In this defect an atom is transferred from a lattice site to an interstitial site, a position which is not occupied by any atom. A Frenkel defect, therefore, has two parts; a vacancy and an interstitial atom. If the interstitial atom does not fall back into its own vacancy, the interstitial atom, the vacancy or both may diffuse away from their points of creation. Then, we get a free interstitial and a free vacancy. In this way it is after various degrees of dissociation that the two components of Frenkel defect become independent of each other.

Let, the energy required to displace an atom from a normal site to an interstitial position in a perfect crystal be E_i . Let, there be N atoms in the crystal and N' interstitial atoms within it.

The number of ways in which n Frenkel defects can be formed is given by,

$$W = \frac{N!}{(N-n)! n!} \cdot \frac{N'!}{(N'-n)! n!} \quad \dots \dots \dots \text{--- (1)}$$

The change in the Helmholtz free energy produced due to the formation of a Frenkel defect is given by,

$$F = U - TS \text{ ----- (2)}$$

where, U - is the internal energy of the crystal and S is its entropy.

If n - be the number of vacancy pairs, then the total increase in the energy due to production of the vacancies is given by,

$$U = nE_i \text{ ----- (3)}$$

The entropy is given by Boltzmann's statistical relation,

$$S = k_B \log w \text{ ----- (4)}$$

using equations (1), (3) and (4) into eqn (2), we obtain

$$\begin{aligned} F &= nE_i - k_B T \log \left[\frac{N!}{(N-n)! n!} \cdot \frac{N'!}{(N'-n)! n!} \right] \\ &= nE_i - k_B T \left[\log \frac{N!}{(N-n)! n!} + \log \frac{N'!}{(N'-n)! n!} \right] \\ &= nE_i - k_B T \left[\log N! - \log (N-n)! - \log n! + \log N'! - \right. \\ &\quad \left. \log (N'-n)! - \log n! \right] \\ &= nE_i - k_B T \left[\log N! + \log N'! - \log (N-n)! - \log (N'-n)! \right. \\ &\quad \left. - 2 \log n! \right] \end{aligned}$$

using stirling's formula, $\log x! \approx x \log x - x$

Then,

$$\begin{aligned} F &= nE_i - k_B T \left[N \log N - N + N' \log N' - N' - (N-n) \log (N-n) + (N-n) \right. \\ &\quad \left. - (N'-n) \log (N'-n) + (N'-n) - 2n \log n + 2n \right] \\ &= nE_i - k_B T \left[N \log N + N' \log N' - (N-n) \log (N-n) - (N'-n) \log (N'-n) \right. \\ &\quad \left. - 2n \log n \right] \text{ ----- (5)} \end{aligned}$$

Differentiating equⁿ ⑤ with respect to n , we get,

$$\frac{\partial F}{\partial n} = E_i - k_B T \left[0 + 0 - \left(\frac{N-n}{N-n} \right) (-1) - \log(N-n) (-1) - (N'-n) \frac{1}{(N'-n)} (-1) \right.$$

$$\left. - \log(N'-n) (-1) - 2 \log n - 2 \right]$$

$$= E_i - k_B T [1 + \log(N'-n) + 1 + \log(N-n) - 2 \log n - 2]$$

$$= E_i - k_B T [\log(N-n) + \log(N'-n) - 2 \log n]$$

$$= E_i - k_B T \left[\log \frac{(N-n)(N'-n)}{n^2} \right]$$

since, at equilibrium the energy is constant, so $\frac{\partial F}{\partial n} = 0$, then we have,

$$E_i = k_B T \left[\log \frac{(N-n)(N'-n)}{n^2} \right]$$

If $n \ll N$ and $n \ll N'$, then we can write,

$$N-n \simeq N \quad \text{and} \quad N'-n \simeq N'$$

$$\therefore E_i = k_B T \log \frac{NN'}{n^2}$$

$$\Rightarrow \log \left[\frac{(NN')^{1/2}}{n} \right]^2 = \frac{E_i}{k_B T}$$

$$\Rightarrow 2 \log \frac{(NN')^{1/2}}{n} = \frac{E_i}{k_B T}$$

$$\Rightarrow \log \frac{(NN')^{1/2}}{n} = \frac{E_i}{2k_B T}$$

$$\Rightarrow \frac{(NN')^{1/2}}{n} = \exp \left(\frac{E_i}{2k_B T} \right)$$

$$\Rightarrow n = (NN')^{1/2} \exp \left(-\frac{E_i}{2k_B T} \right)$$

when, $(NN')^{1/2}$ is constant, then

$$n \simeq \text{const} \exp \left(-\frac{E_i}{2k_B T} \right).$$

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State Fick's law. show that, the diffusion constant is given by,

$$D = (va^2) \exp\left(-\frac{E}{k_B T}\right)$$

where the terms carry usual meaning.

⇒ when there is a concentration gradient of impurity atoms or vacancies in a solid, there will be a flux of these through the solid.

In equilibrium, the impurities or vacancies will be distributed uniformly. The net flux $\bar{J}_N$ of atoms of one species in a solid is related to the gradient of the concentration N of this species by Fick's law,

$$\bar{J}_N = -D \text{grad } N \quad \text{--- (1)}$$

where, $\bar{J}_N$ - is the number of atoms crossing unit area in unit time and D - is the diffusion constant.

The diffusion constant is often found, to vary with temperature as,

$$D = D_0 \exp\left(-\frac{E}{k_B T}\right) \quad \text{--- (2)}$$

where, E - is the activation energy for the process.

we consider, the diffusion of impurity atom in interstitial sites identical results apply to the diffusion of vacancies. If the barrier height E , the atom will have sufficiently thermal energy to pass over the barrier only a fraction $\exp\left(-\frac{E}{k_B T}\right)$ of the time.

If ν is the characteristic atomic vibrational frequency then the probability P that sometime

during one second is,

$$P \approx \nu \exp\left(-\frac{E}{k_B T}\right) \text{ --- --- --- --- --- (3)}$$

The quantity P is called jump frequency.

Let us consider two parallel planes of impurity atoms in the interstitial sites. The planes are separated by lattice constant a . There are ' s ' impurity atoms on one plane and $(s + a \frac{ds}{dx})$ on the other. The net number of atoms crossing between the plane in one second is $\approx -Pa \frac{ds}{dx}$.

If N is the total concentration of impurity atoms, then $s = aN$, per unit area of a plane.

The diffusion may now be written as,

$$\begin{aligned} J_N &= -Pa \frac{ds}{dx} \\ &= -Pa^2 \frac{dN}{dx} \quad [\because s = aN] \\ &= -\nu a^2 \exp\left(-\frac{E}{k_B T}\right) \text{grad } N \text{ --- --- --- --- --- (4)} \end{aligned}$$

Now, comparing equⁿ (1) and (4), we get,

$$\begin{aligned} D &= \nu a^2 \exp\left(-\frac{E}{k_B T}\right) \\ &= D_0 \exp\left(-\frac{E}{k_B T}\right) \text{ --- --- --- --- --- (5)} \end{aligned}$$

where, $D_0 = \nu a^2$

[If the impurity changed, we may find the ionic mobility μ and conductivity σ from the diffusivity

by using the Einstein relation,

$$k_B T \mu = q D$$

$$\Rightarrow \mu = \left(\frac{q^2 \nu a^2}{k_B T} \right) \exp \left(- \frac{E}{k_B T} \right) \quad \text{--- (6)}$$

$$\text{and } \sigma = N q \mu = \left(\frac{N q^2 \nu a^2}{k_B T} \right) \exp \left(- \frac{E}{k_B T} \right) \quad \text{--- (7)}$$

where, N - is the concentration of impurity ions of charge q .

97, 9903 # what is colour centre? Explain the formation of F-centre in an alkali halide crystal with the help of a suitable physical model.

96. ⇒ Colour centre:-

we know, a perfect NaCl crystal should not absorb visible light and so it should be perfectly transparent. This leads us to that the colouration of crystals is due to defects in the crystal.

It is known that, insulators have large energy gaps and they are transparent to visible light. However, due to different reasons, absorption band is in the visible near ultraviolet and near infrared regions in crystals like KCl, NaCl, LiF and other alkali halides.

If the absorption band is in the visible region and the band is quite narrow, it gives a characteristic colour to the crystal. When the crystal gets coloured

it is said to have colour centres. Thus a colour centre is a lattice defect which absorbs light.

It is possible to colour the crystals in number of different ways as -

1. By the addition of suitable chemical impurities.
2. By heating the crystal in the alkali metal vapour and then cooling it quickly, e.g., LiF heated in Li-vapour colours it pink, an excess of Na in NaCl makes the crystal yellow.

3. By exposing crystals to high energy radiation like x-ray, γ -ray or by bombarding them with energetic electrons or neutrons. colour centre

Formation of F-centre :-

The formation of F-centre was suggested by De Boer and was further developed by Mott and Gurney.

This model is based on the non-stoichiometric properties of crystals, i.e., crystals have an excess of one of its constituents.

Let us take NaCl crystal. It can be coloured by heating it in an atmosphere of sodium vapour and then cooled it quickly. The excess Na atoms absorbed from the vapour split up into electrons and positive ions in the crystal. This results Cl^- vacancies. The valence electron of the alkali atom is not bound to the

atom, it diffuses into the crystal and becomes bound to a vacant negative ion site at F. It just traps an electron in order to maintain local charge neutrality.

The excess electron capture in this way at a negative ion vacancy in an alkali halide crystal is called an F-centre.

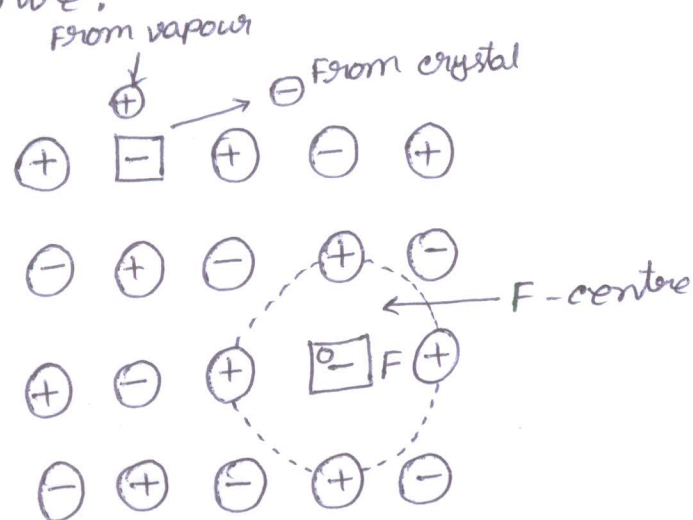


Figure:- Formation of F-centre in crystals.

95, 98, 01 # Explain the phenomena of order-disorder transformation in an alloy.

⇒

99.03 # what are edge and screw dislocations? Derive an expression for the energy of screw dislocation.

⇒ Edge dislocation:

It occurs when the periodicity of the atomic lattice array is interrupted along certain directions in a crystal. An edge dislocation is formed by a missing half-plane of atoms.

Screw dislocation:

It is a type of dislocation in which the dislocation line is parallel to the slip direction and it has a row of atoms about which a normal crystallographic plane appear to be spiral.

Expression:

Let us have a cylindrical shell of a material surrounding an axial screw dislocation. Let the radius of the shell be r and the thickness dr . The circumference of the shell is $2\pi r$ and let it be sheared by an amount b , so that, the shear strain,

$$e = \frac{b}{2\pi r} \text{ --- (1)}$$

and the corresponding shear stress in the good region is,

$$\tau = G \cdot e = \frac{G \cdot b}{2\pi r} \text{ --- (2)}$$

where, G is the shear modulus of rigidity of the material.

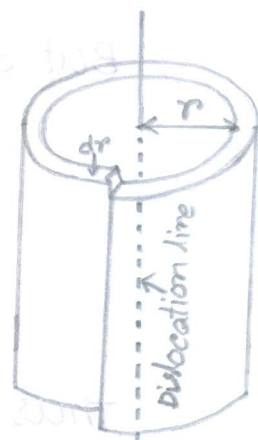


Fig:

A distribution of forces is exerted over the surface of the cut for producing a displacement b and the work done by the forces to do it gives the energy E_s of the screw dislocation. Hence,

$$E_s = \int F b dA \quad \text{--- (3)}$$

where, F - is the average force per unit area at a point on the surface during the displacement and the integral extends over the surface area of the cut. The average value is to be taken because the force at a point builds up linearly from zero to a maximum value as the displacement is produced. Thus the average force is half the final value when the displacement is b , i.e., $F = \langle \tau \rangle = \frac{1}{2} \tau$.

$$\therefore F = \frac{Gb}{4\pi r} \quad \text{--- (4)}$$

Putting it in equⁿ (3), we get,

$$E_s = \int \frac{Gb^2}{4\pi r} dA$$

But $dA = dz \cdot dr$ and for a dislocation of length l , we have,

$$\begin{aligned} E_s &= \int_0^R \int_{r_0}^R \frac{Gb^2}{4\pi r} dz dr \\ &= \frac{Gb^2}{4\pi} \log \frac{R}{r_0} \quad \text{--- (5)} \end{aligned}$$

Thus, the total elastic energy per unit length of a screw dislocation is given by,

$$E_s = \frac{Gb^2}{4\pi} \log \frac{R}{r_0} \quad \text{--- (6)}$$

where, R and r_0 are the upper and lower limits of r .

99.03 # Discuss the role of screw dislocation in the growth of single crystal.

⇒ The theory of growth of perfect crystals predicts that an oversaturation of about 25 to 50% is required to nucleate at sufficient frequency. Actually, an oversaturation of about 1% is sufficient for the growth of a crystal. Thus, there is a large disagreement between theory and practice.

This difficulty is removed assuming a step growth be also continuous and there may be no necessity of formation of a new nucleus every time. This is achieved if a screw dislocation be present in the crystal because the crystal will grow in a spiral form at the edge of the discontinuity.

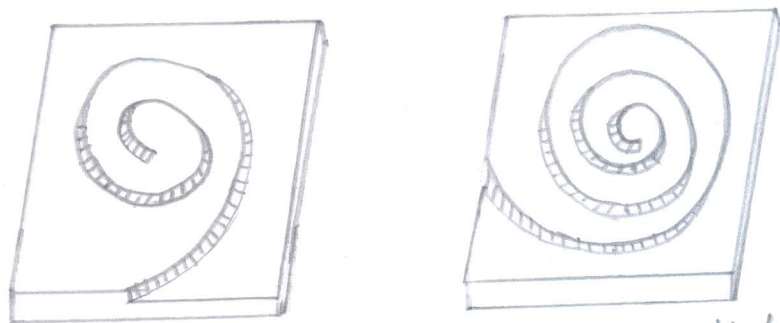


Fig:- crystal growth by screw dislocation

The continuous development of a spiral step produced by the intersection of a screw dislocation with the surface of a crystal is shown in fig. The presence of the screw dislocation of a surface prevents nucleation of a completed surface. The calculated rates of growth on this model are in agreement with experimental results.

99.03 # calculate the elastic energy of line imperfections stored in an aluminium crystal. [Given: shear modulus = 25.94 G N m^{-2} ; dislocation density = 10^{10} m^{-2} ; $a = 4.05 \text{ \AA}$]



